Reusable Change Signal: An Outcome-Blind Audit of Annual Earth Embeddings

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Abstract

Precomputed annual Earth embeddings offer a reusable measure of landscape change without task-specific training. We evaluate whether AlphaEarth embedding displacement tracks mapped impacts and adds value beyond specialized optical indices. In an outcome-blind cohort of ten Monitoring Trends in Burn Severity (MTBS) fires, displacement recovered substantial within-fire severity ordering (median Spearman $\rho = 0.712$), but showed no consistent advantage over a prespecified Sentinel-2 baseline based on burn-ratio and vegetation-index change. A second embedding product, TESSERA, showed a similar pattern. Across eight development cases involving lava emplacement, forest loss, and drought, AlphaEarth displacement was positively associated with every mapped reference, although effect sizes varied. These findings support annual embeddings as portable change representations, though not as replacements for task-specific measures or operational hazard detectors.

1 Introduction

AlphaEarth Foundations (AEF) generates annual 64-dimensional embeddings for 10 m pixels across terrestrial and coastal areas, with data available from 2017 to 2025 [Brown et al., 2025, Google, 2025]. Each embedding summarizes surface conditions and their temporal trajectory at and around a pixel over a year [Brown et al., 2025]. Year-to-year AEF change can be measured by comparing the same pixel's embedding across years, but attributing that change to a specific cause is more difficult for two reasons: (1) annual embeddings summarize a full year, so short events may be represented alongside pre- and post-event conditions; and (2) an external map used to explain AEF changes may be inaccurate in the study area [Comber et al., 2012, Olofsson et al., 2014] or have little local variation (e.g., binary extent rather than graded severity) [Key and Benson, 2006], limiting its ability to explain differences in embedding change.

This study addresses two related but distinct questions. First, how closely does annual AlphaEarth embedding displacement track disturbance patterns shown in an external reference map? Here, "track" refers only to an association across spatial cells and does not imply disturbance detection, causal attribution, or ground-truth accuracy. Second, does embedding displacement provide information beyond changes in conventional spectral indices? The first question evaluates **representation fidelity**, meaning how well AlphaEarth reflects the mapped disturbance pattern, while the second evaluates its **incremental value**, or whether it adds information beyond simpler spectral indices. We contribute (i) an evaluation design that separates reference-aligned representation fidelity from incremental value over a task-aligned baseline; (ii) an outcome-blind, frozen AlphaEarth test on ten MTBS fires; and (iii) an eight-case development audit of the same AlphaEarth score across lava emplacement, forest loss, and drought. The held-out and development evidence have distinct roles: the former tests representation fidelity and added value, while the latter provides preliminary evidence of reuse without establishing cross-process generalization.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: 2nd Workshop on Advances in Representation Learning for Earth Observation.

2 Related work

Recent work has evaluated geospatial embeddings in different ways. AlphaEarth has been evaluated across a range of mapping tasks, including classification, regression, and change detection [Brown et al., 2025]. TESSERA (Temporal Embeddings of Surface Spectra for Earth Representation and Analysis) provides open annual 128-dimensional Sentinel-1/2 embeddings and is our direct representation comparator [Feng et al., 2026]. NeurEO studies scale-dependent organization, Rahman environment-aligned coordinates, and Benavides-Martinez et al. task-specific subsets; together, these adjacent studies establish decodability and redundancy of state embeddings, not universal process-change axes or added value over process-specific optical change [van der Plas et al., 2025, Rahman, 2026, Benavides-Martínez et al., 2026]. In contrast, we evaluate unsupervised year-to-year embedding displacement against external disturbance maps, treated as reference products rather than ground truth.

Our event-specific references are USGS lava extents for Kilauea and Mauna Loa and CEMS extent for La Palma [Zoeller et al., 2020, Copernicus Emergency Management Service, 2021, Zoeller et al., 2024], Hansen annual forest loss [Hansen et al., 2013], the multi-input U.S. Drought Monitor [Svoboda et al., 2002], and USFS/USGS MTBS thematic severity, derived from Landsat imagery and dNBR [Eidenshink et al., 2007].

3 Method and evaluation protocol

Annual displacement. For model m, let $u_{m,p}^{(y)}$ denote its unit-normalized annual embedding at pixel p in year y; we compute

$$d_{m,p}^2 = \|u_{m,p}^{(\mathrm{post})} - u_{m,p}^{(\mathrm{pre})}\|_2^2 = 2 - 2\cos(u_{m,p}^{(\mathrm{post})}, u_{m,p}^{(\mathrm{pre})}).$$

These values are averaged within H3 hexagonal grid cells [H3 Project, 2026]; we call the resulting AlphaEarth quantity embedding change squared (ECS). ECS is a descriptive measure of change magnitude and is not intended as a hazard detector. Our representation comparator, TESSERA v1.0 (vultr manifest), provides int8 vectors that are dequantized and l_2-normalized per pixel before the same displacement and H3 averaging on the frozen fire cells. Because model scales are not directly comparable, each model's displacement is instead correlated by Spearman rank with the same event reference; cross-model comparisons use those within-fire coefficients.

Outcome-blind held-out MTBS fire test. MTBS analysts derive dNBR from pre- and post-fire Landsat imagery and interpret it with the imagery, ecological context, ancillary evidence, and, where available, field observations such as Composite Burn Index plots to set event-specific severity thresholds [Eidenshink et al., 2007, Picotte et al., 2020]. We therefore use MTBS as the study's standardized cohort-wide burn-severity reference. Before AlphaEarth access, a frozen metadata snapshot deterministically selected ten U.S. fires plus ordered reserves across years, regions, sizes, and dominant land-cover strata. Official boundaries and 30 m MTBS thematic rasters define the reference within each H3-r8 cell (resolution 8, $\sim 0.74\,\mathrm{km}^2$ mean area) as the fraction of classes 3–4 (moderate/high severity) among classes 1–5.[1] Class 2 is valid low severity, contributing to the denominator but not the numerator. Background/NoData and non-processing areas are excluded [Eidenshink et al., 2007]. Cell centers must lie within the event boundary, at least 80% of each cell must have valid MTBS support, and at least 200 cells must remain. The exact H3 population is frozen before AlphaEarth extraction. Four County had a constant target and, before AlphaEarth access, was replaced by the next prespecified reserve, Walker.

For an event in year y, AlphaEarth, TESSERA, and Sentinel-2 use full calendar years $y-1$ and $y+1$. The optical comparator median-composites cloud-masked `S2_SR_HARMONIZED` reflectance [European Space Agency, 2021] and computes two spectral indices, dNBR and dNDVI, per H3 cell, with positive values indicating post-event decrease (Appendix B).

Event-level estimands. Within-event Spearman correlation (ρ) summarizes representation fidelity: whether embedding displacement recovers MTBS-aligned severity ordering. It is not an equivalence

[1] MTBS thematic classes: 1 unburned to low, 2 low, 3 moderate, 4 high severity, 5 increased greenness [Eidenshink et al., 2007].

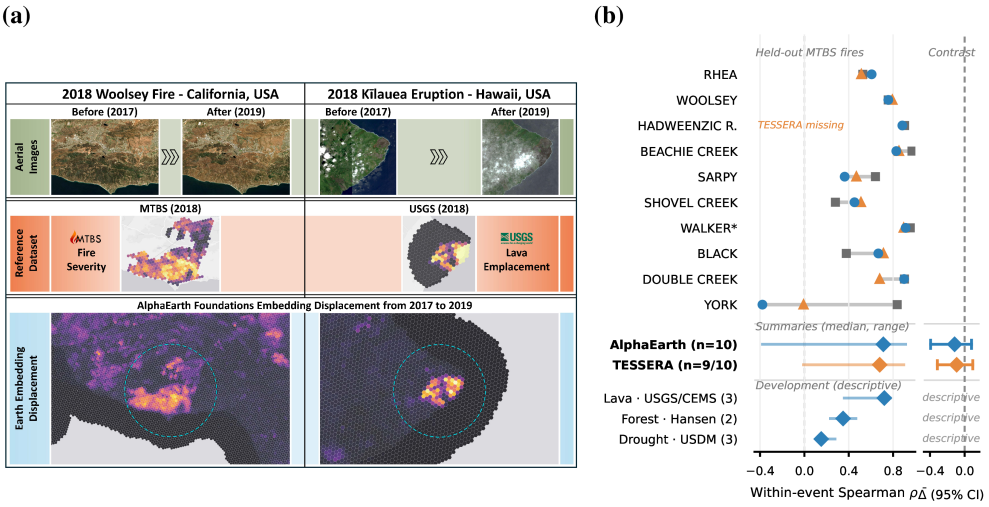


Figure 1: **Fidelity illustration and frozen evidence summary.** (a) One held-out fire and one development case, both 2018 events with displacement bracketed 2017 → 2019: the Woolsey Fire against MTBS severity (within-event $\rho = 0.757$) and the Kilauea eruption against USGS lava extent (within-event $\rho = 0.774$). Rows show before/after imagery, the official reference on the common H3 grid, and AlphaEarth displacement (ECS). Each official reference is task-specific, whereas the AlphaEarth displacement shown is the same unsupervised score in both cases—and in every case of panel (b). (b) Graphical summary of Table 1 (Appendix A): per-event associations for the ten held-out fires (blue circles AlphaEarth, gray squares best-of-two optical, orange triangles TESSERA; Walker* is the prespecified reserve; one official TESSERA year tile is missing), model and development-family summaries (diamonds medians, lines ranges), and mean incremental contrasts with 95% event-bootstrap intervals (dashed line at zero). Map colors are normalized per panel.

criterion. For fire e, the frozen primary statistic is

$$\Delta_e^{\mathrm{A}} = \rho(\mathrm{ECS}, \mathrm{MTBS}) - \max\{\rho(\mathrm{dNBR}, \mathrm{MTBS}), \rho(\mathrm{dNDVI}, \mathrm{MTBS})\}.$$

This statistic measures standalone advantage and does not assess whether displacement complements the optical indices. The prespecified Δ_e^{T} substitutes normalized TESSERA displacement on the same frozen cells. We report unweighted event means and 95% percentile intervals from 100,000 event-bootstrap resamples. Each fire, not its autocorrelated H3 cells, is one inferential unit [Clifford et al., 1989]; no cell-level p-values are used. Selecting the stronger of dNBR and dNDVI deliberately favors the optical baseline; the directional alternative is $\bar{\Delta}^{\mathrm{A}} > 0$. Spearman association is primary because the MTBS target derives from thematic classes. All correlations rank severity within official perimeters, with no unaffected outside-perimeter cells.

Development audit and temporal support. Mean ECS and reference values are reduced to predefined H3 cells; only finite cells valid in both layers are retained. Each eruption is bracketed by the preceding and following annual embeddings: 2017 → 2019 for Kilauea, 2020 → 2022 for La Palma, and 2021 → 2023 for Mauna Loa. Lava support uses WorldCover non-water pixels plus mapped lava; La Palma also restricts non-lava support to the CEMS analyzed footprint [Zanaga et al., 2021]. Two 2021 forest comparisons multiply Hansen `lossyear` indicators by year-2000 canopy cover. Three drought comparisons use absolute change between fixed April–October mean USDM scores, avoiding event-specific peak-month selection. We score weekly USDM categories as none = 0 and D0–D4 = 1–5; absolute change treats intensification and recovery alike. Development cases were not screened by correlation magnitude or sign.

4 Results

Held-out fidelity and incremental value. A simple AlphaEarth displacement score recovered substantial but heterogeneous MTBS-aligned severity ordering (Figure 1). Nine of ten fires had positive associations, with median $\rho = 0.712$ (exact two-sided event-level sign test, $p = 0.021$). AlphaEarth exceeded the optical comparator in four of ten. The frozen mean endpoint was $\bar{\Delta}^{\mathrm{A}} = -0.116$ (95% interval $[-0.393, 0.079]$); median $\Delta_e^{\mathrm{A}} = -0.012$. Thus, the study supports representation fidelity but establishes superiority for neither AlphaEarth nor the optical comparator. The secondary analysis with the representation comparator, TESSERA, was prespecified on the frozen fire populations. Nine events were evaluable; the official v1.0 (vultr) manifest lacked one required year–tile object for the tenth, which remained in the AlphaEarth cohort as TESSERA-missing. TESSERA exceeded the optical comparator in three of nine events; $\bar{\Delta}^{\mathrm{T}} = -0.092$ (95% interval $[-0.315, 0.094]$). The descriptive paired mean AlphaEarth-minus-TESSERA ρ difference was -0.034 (95% interval $[-0.138, 0.059]$). Neither interval establishes a consistent advantage.

Development evidence. The family summaries include all predefined cases without outcome screening. All eight associations are positive, with family medians of 0.721 (lava), 0.349 (forest loss), and 0.152 (drought) (Table 1, Appendix A), providing early evidence that the same unsupervised change score may be reusable across distinct environmental processes. Post hoc spatial sensitivity remained positive at the primary block scale in all eight cases but mixed or unassessable at coarser scales (Appendix C). These heterogeneous development cases do not establish general cross-process performance or a family-level effect.

These analyses evaluate direct reuse of precomputed displacement; we did not measure label efficiency or train downstream models. Whether displacement adds information conditional on the optical indices, for example, through partial rank correlation on the frozen cells, is the natural next analysis and was not prespecified here. Precomputed embeddings supply one change representation that transfers across processes, whereas the matched optical indices remained competitive within the fire cohort; we therefore read the cross-process results as evidence of portability rather than of task-level advantage.

5 Validity, omissions, and limitations

The development audit is deliberately heterogeneous and cannot estimate a population-level effect. Only the fire cohort was selected outcome-blind and held out before AlphaEarth extraction, and ten events give a wide bootstrap interval. One retained event (YORK) has $\Delta_e^{\mathrm{A}} = -1.215$; omitting it would give a descriptive mean of 0.006. We retain the event and keep the frozen endpoint unchanged (Appendix D). The MTBS comparison is not independent optical validation: analysts derive thematic classes largely by thresholding Landsat dNBR, with interpretation and occasional ancillary evidence [Eidenshink et al., 2007]. Our Sentinel-2 dNBR is not the same image product, but it targets the same spectral construct. We therefore interpret within-event ρ as criterion-aligned fidelity, and $\bar{\Delta}^{\mathrm{A}}$ and $\bar{\Delta}^{\mathrm{T}}$ as added value relative to a product-aligned baseline, not as a general embedding-versus-imagery verdict.

External maps have omissions and class semantics; valid-support exclusion changes the target population; and annual embeddings may respond to phenology, management, sensor/compositing effects, or unrelated change. H3 averaging smooths sub-cell structure. Because MTBS cells are sampled only inside mapped perimeters, the study ranks severity within them and does not address fire detection, localization, or behavior on unaffected landscapes. Artifact and replay details appear in Appendix E.

6 Conclusion

A simple difference between readily available annual embeddings recovered substantial MTBS-aligned severity ordering in most held-out fires, with no consistent standalone advantage over target-aligned spectral indices for either AlphaEarth or TESSERA. Positive but variable development associations across lava emplacement, forest loss, and drought support annual embeddings as broadly reusable sources of mapped-change information; on this evidence they do not replace specialized indices or operational hazard detectors.

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A Frozen evidence summary

Table 1: Frozen evidence summary. Median ρ summarizes representation fidelity. Held-out contrasts compare each embedding with the prespecified best-of-two optical comparator, then average equally over events. Intervals resample events. Development rows summarize predefined non-fire cases by family.

Study	Reference	Events	Fidelity: median ρ (range)	Incremental contrast (95% interval)
AlphaEarth held-out	MTBS severity	10	$0.712\ (-0.378, 0.913)$	$\bar{\Delta}^{\mathrm{A}} = -0.116\ (-0.393, 0.079)$
TESSERA secondary	MTBS severity	9/10	$0.679\ (-0.008, 0.896)$	$\bar{\Delta}^{\mathrm{T}} = -0.092\ (-0.315, 0.094)$
Lava development	USGS/CEMS extent	3	$0.721\ (0.360, 0.774)$	descriptive
Forest development	Hansen loss	2	$0.349\ (0.235, 0.464)$	descriptive
Drought development	USDM change	3	$0.152\ (0.126, 0.276)$	descriptive

B Sentinel-2 comparator details

B4 (red), B8 (near-infrared), and B12 (shortwave infrared) are median-composited after filtering scenes to at most 70% reported cloud cover and masking cloud, shadow, cirrus, snow, and invalid pixels; each pixel requires at least two clear observations per year. NDVI summarizes near-infrared/red vegetation contrast, while NBR uses near-infrared/shortwave-infrared contrast sensitive to burn and moisture change. We define $\mathrm{NDVI} = (B8 - B4)/(B8 + B4)$, $\mathrm{NBR} = (B8 - B12)/(B8 + B12)$, $\mathrm{dNBR} = \mathrm{NBR}_{\mathrm{pre}} - \mathrm{NBR}_{\mathrm{post}}$, and $\mathrm{dNDVI} = \mathrm{NDVI}_{\mathrm{pre}} - \mathrm{NDVI}_{\mathrm{post}}$. Positive values indicate post-event decreases. Both indices are averaged over valid pixels within each H3-r8 cell.

C Spatial-sensitivity details

A frozen post hoc spatial sensitivity retained positive primary-scale block correlations in all eight cases (equal-event median Spearman correlation 0.424); five passed Holm-adjusted block-label permutation tests, while coarsest-scale results were mixed or unassessable in six cases. Free block-label permutation does not guarantee exchangeability across broad spatial gradients; this sensitivity is supplementary and scale-dependent. Because these are heterogeneous development cases, they do not establish general cross-process performance or a family-level effect.

D YORK target diagnostic

The ten AlphaEarth coefficients span -0.378 to 0.913 over 6,864 frozen event-cells. YORK (2023, Mojave shrub/sparse; $n = 410$) passed every QA gate but has $\rho_{\mathrm{ECS}} = -0.378$ for the frozen moderate/high target. A labeled post hoc diagnostic found that 84.0% of mapped pixels were MTBS class 2: ECS was near zero for an ordinal target ($\rho = -0.038$) but positive for any-severity fraction ($\rho = 0.344$), while remaining below matched optical change in both. This is consistent with sensitivity to low-severity extent rather than severity ordering; the frozen endpoint is unchanged.

E Reproducibility

The artifact includes canonical pre-outcome protocols and hashes, failed attempts and reserve ordering, immutable H3 populations, per-event metrics, official-reference replay, and code for the event bootstrap. All weak, negative, and missing cases are retained. The artifact is available at `https://github.com/whotank/alphaearth-disaster-pipeline`.